

The relative position of lines in space

Parallelism revisited where a third case exists — and the two theorems that make it usable.

LESSON 19–20

2 × 40 MIN

GEOMETRY 10, §9

EXTENSION

LESSON CLOCK

10'

20'

22'

20'

16'

2

The definition, restated	10 min
One parallel through a point	20 min
Transitivity	22 min
Respectively parallel sides	20 min
Practice	16 min
Homework	2 min

LEARNING OBJECTIVES

- State the definition of parallel lines in space.
- Prove that through a point not on a line there is exactly one parallel to it.
- Apply the transitivity of parallelism in space.
- Prove the theorem on the angle between lines with respectively parallel sides.

TEXTBOOK

National	pp. 83–92
Cambridge	Core & Extended, pp. 277–281
Workbook	Exercise 9.1

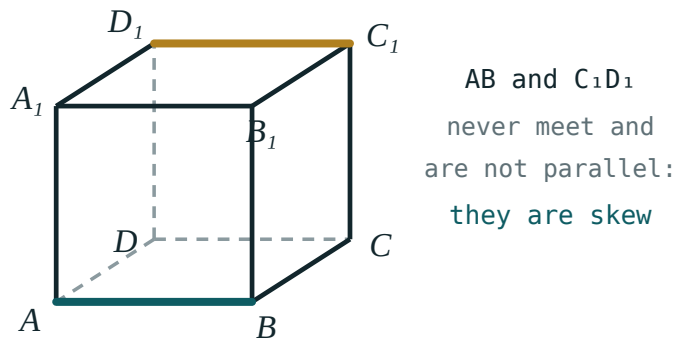
01

Explanation

The definition needs one more word

PARALLEL IN SPACE

Two lines are **parallel** when they lie in **one plane** and do not meet. Without the first condition the definition would also admit skew lines, which never meet either.



*Skew lines never meet — and no plane holds both.
That is exactly what parallelism excludes.*

Everything else about parallel lines carries over from the plane unchanged. What must be re-proved is that they still **exist and are unique** once the extra dimension is available.

One parallel through a point

Theorem. Through a point M not on a line a there passes exactly one line parallel to a .

Proof. The line a and the point M determine a unique plane α (Theorem 1 of Lesson 4–5). Inside α the plane axiom gives exactly one parallel through M . Any parallel to a through M must be coplanar with a by the definition, and that plane contains a and M , so it is α . Hence there is no other. ■

THE PATTERN OF EVERY PROOF IN THIS CHAPTER

Reduce the space problem to a plane problem by naming the right plane, then use plane geometry. Naming the plane is the whole of the work.

Transitivity

Theorem. If $a \parallel b$ and $b \parallel c$ then $a \parallel c$.

In the plane this is obvious. In space it is not, because a and c need not obviously share a plane — and until that is proved, they might be skew. The proof constructs the plane and shows they cannot meet.

PERPENDICULARITY IS NOT TRANSITIVE

In space, $a \perp b$ and $b \perp c$ does **not** give $a \perp c$. Take three edges of a cube meeting at one vertex: any two are perpendicular, so the conclusion would be false. Parallelism is transitive; perpendicularity

is not.

Respectively parallel sides

Theorem. If two angles in space have respectively parallel sides, they are either equal or supplementary.

This is what allows the angle between skew lines to be well defined: translating a line does not change the angle it makes with anything, because the translated line is parallel to the original and its sides are respectively parallel.

translate b to meet $a \implies$ the angle is unchanged

Which of “equal” or “supplementary” occurs depends on the directions chosen; the convention is to take the acute one.

02 Worked examples

EXAMPLE 1 In cube $ABCD A_1 B_1 C_1 D_1$, prove $AB \parallel D_1 C_1$.

① $AB \parallel DC$
Opposite sides of a square face.

② $DC \parallel D_1 C_1$
Opposite sides of face $DCC_1 D_1$.

③ By transitivity.

ANSWER

$AB \parallel D_1 C_1$

EXAMPLE 2 Give three lines in a cube with $a \perp b$ and $b \perp c$ but a not perpendicular to c .

- ① Take $a = AB$, $b = AA_1$, $c = D_1C_1$.
- ② $AB \perp AA_1$
Adjacent edges of a face.
- ③ $AA_1 \perp D_1C_1$
A vertical edge against a horizontal one.
- ④ But $AB \parallel D_1C_1$, so the angle between them is 0° .
Not perpendicular.

ANSWER

AB, AA_1, D_1C_1 — the first and last are parallel

EXAMPLE 3 M is a point not on line a . How many lines through M are skew to a ?

- ① Only one is parallel.
- ② Those in the plane α meet a or are that parallel.
- ③ Every other line through M leaves α .

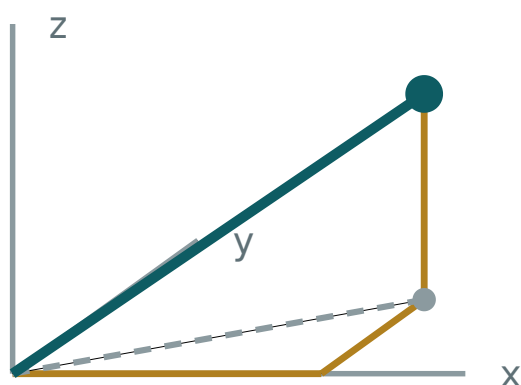
ANSWER

Infinitely many

03 Interactive model

Hold two pencils parallel, then move one out of the plane to show what the definition excludes.

Lines through a point



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Parallel lines in space	Fazoda parallel chiziqlar	Параллельные прямые в пространстве
Coplanar lines	Bir tekislikdagi chiziqlar	Компланарные прямые
Transitivity	Tranzitivlik	Транзитивность
Respectively parallel sides	Mos parallel tomonlar	Соответственно параллельные стороны
Unique parallel	Yagona parallel	Единственная параллельная
Direction of a line	Chiziq yo'nalishi	Направление прямой
Angle between lines	Chiziqlar orasidagi burchak	Угол между прямыми
Auxiliary plane	Yordamchi tekislik	Вспомогательная плоскость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Parallel lines in space must:

not meet

lie in one plane and not meet

be equal

be perpendicular to a third

Through a point off a line the number of parallels is:

0

1

2

infinitely many

Parallelism in space is:

not transitive

transitive

symmetric only

undefined

Perpendicularity in space is:

transitive

not transitive

the same as parallelism

undefined

Angles with respectively parallel sides are:

always equal

equal or supplementary

always supplementary

unrelated

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Define parallel lines in space

ANSWER coplanar and never meeting

2. How many parallels through a point off a line?

ANSWER 1

3. Is $AB \parallel D_1C_1$ in a cube?

ANSWER yes

4. Is parallelism transitive in space?

ANSWER yes

5. Is perpendicularity transitive in space?

ANSWER no

6. Name a line skew to AB in a cube

ANSWER CC_1 (or DD_1)

7. Two lines that never meet are parallel?

ANSWER not necessarily — they may be skew

Medium · the standard question

7 PROBLEMS

1. Prove $AB \parallel D_1C_1$ in a cube

ANSWER Transitivity through DC

2. How many lines through a point are skew to a given line?

ANSWER infinitely many

3. If $a \parallel b$ and b meets c , what can a and c be?

ANSWER meeting or skew

4. Angle between AB and C_1D_1 in a cube

ANSWER 0° — they are parallel

5. Angle between AB and B_1C_1

ANSWER 90°

6. Give two lines perpendicular to the same line but not parallel

ANSWER Two edges of a cube meeting a third at one vertex

7. Why is the angle between skew lines well defined?

ANSWER Respectively parallel sides give the same angle

Hard · stretch and reason

7 PROBLEMS

1. Prove transitivity of parallelism in space

ANSWER Construct the plane through a and a point of c , then argue in it

2. Prove: if $a \parallel b$ and a plane meets a , it meets b or contains it

ANSWER Otherwise $b \parallel \alpha$ forces $a \parallel \alpha$

3. In a cube, count the pairs of parallel edges

ANSWER 18 — three directions, $C(4,2) = 6$ each

4. In a cube, count the pairs of intersecting edges

ANSWER 24 — $C(3,2) = 3$ at each of 8 vertices

5. In a cube, count the pairs of skew edges

ANSWER 24

6. Check the three counts against $C(12,2)$

ANSWER $18 + 24 + 24 = 66 \checkmark$

7. Prove the theorem on respectively parallel sides

ANSWER Translate one angle onto the other; the triangles are congruent

07 Homework

Homework — 4 tasks

Every proof must name the plane it works in.

1. Prove that through a point not on a line there is exactly one parallel to it.
2. In cube $ABCD A_1 B_1 C_1 D_1$, prove $AD \parallel B_1 C_1$.
3. Give a counterexample showing perpendicularity in space is not transitive.
4. Explain in three sentences why the definition of parallel lines needs the words “in one plane”.